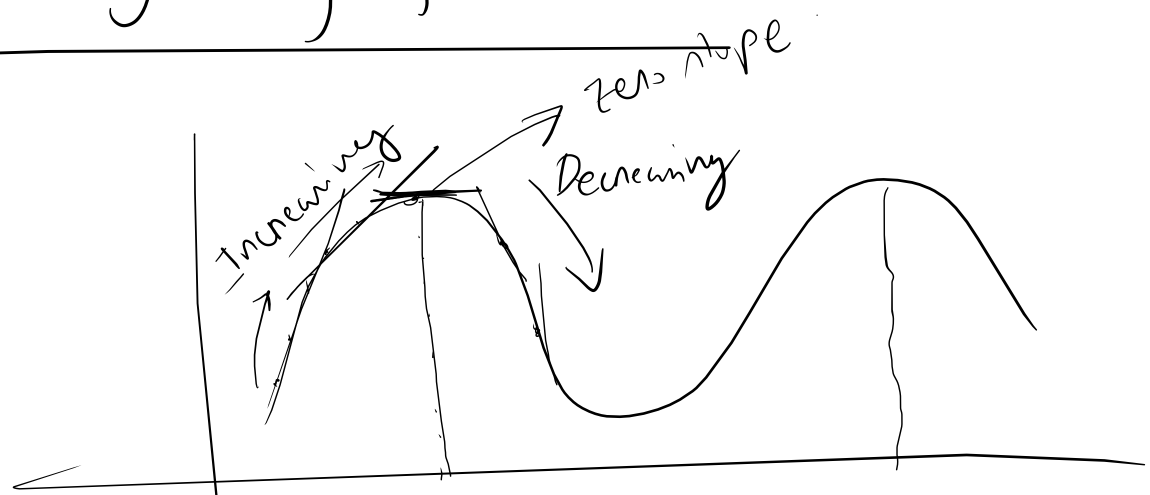
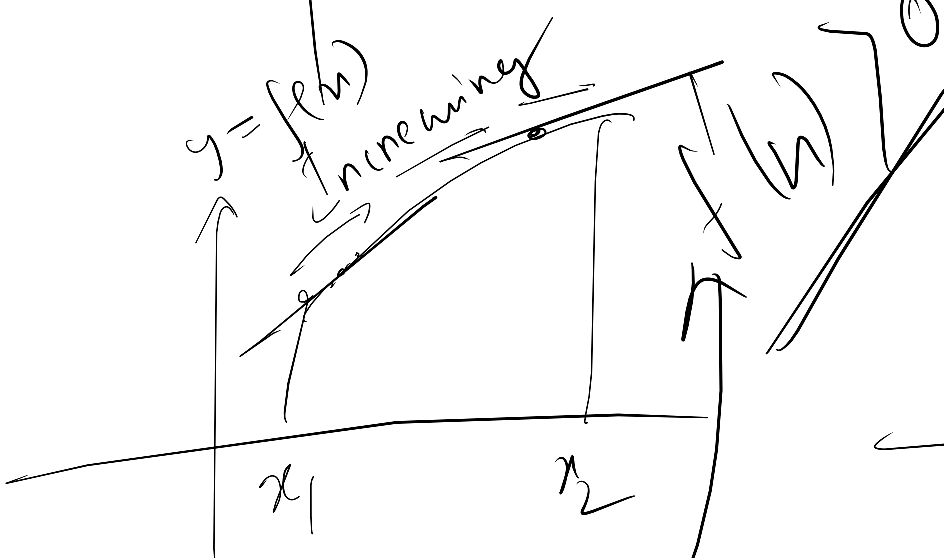
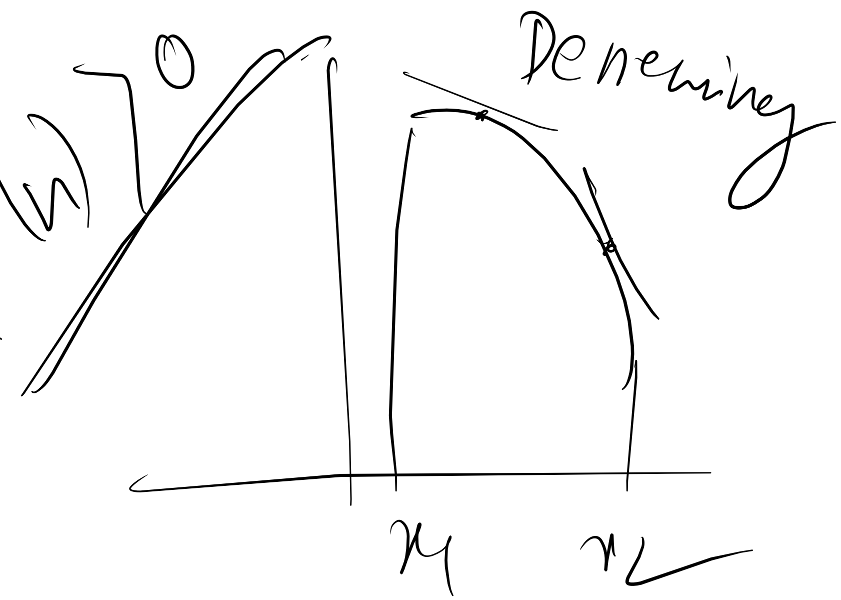
Analysis of function:

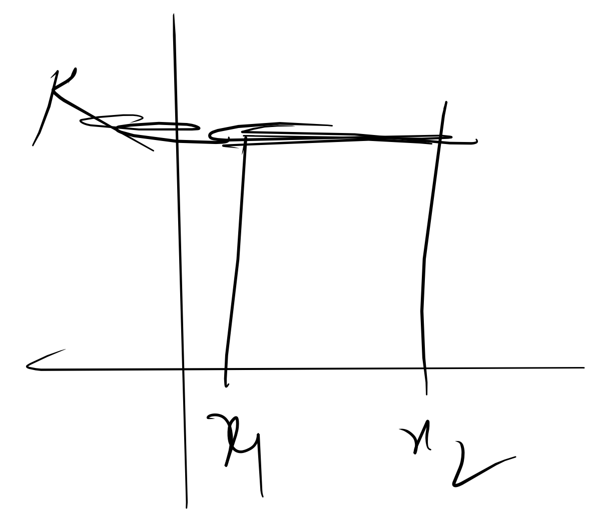




Each tangent line has positive slope.



Each tangent has negative slope



Each tangent has zero slope

Increasing/Decreasing: let f be a function that is continuous on a closed interval $[a,b]$ & differentiable on (a,b).

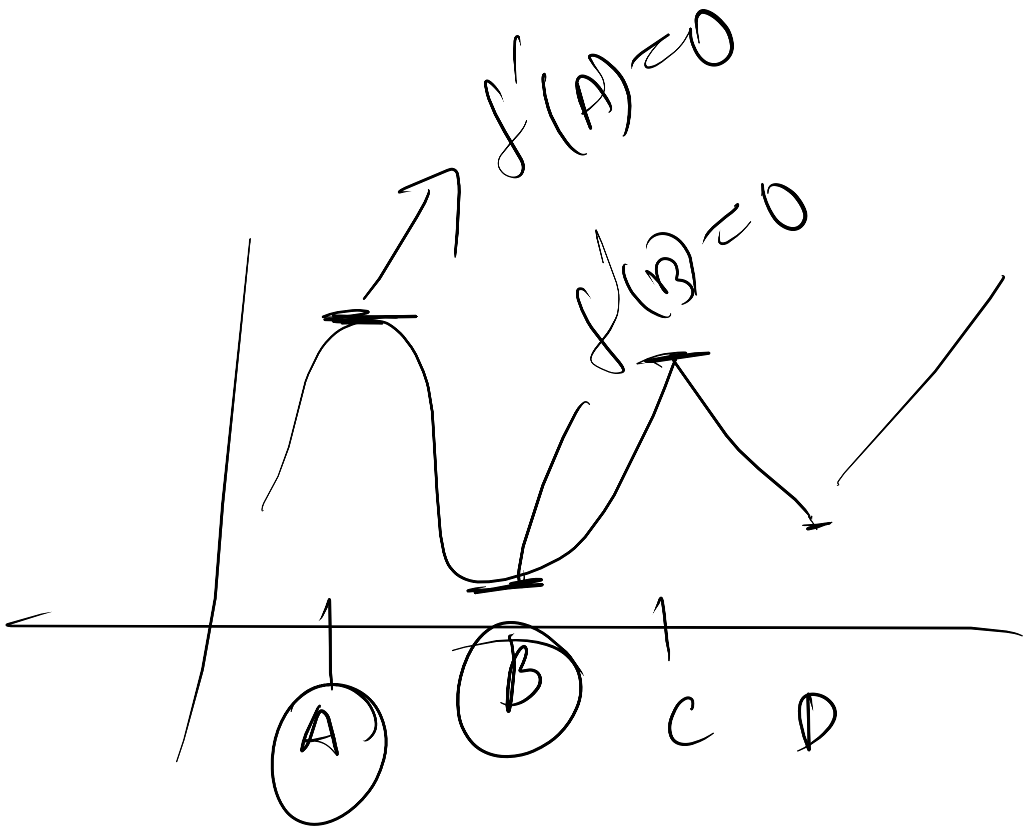
(i) If $f'(x) > 0 \ \forall x \in (a,b)$, $f(x)$ is increasing on $[a,b]$ / (a,b).

(ii) If $f'(x) < 0 \ \forall x \in (a,b)$, $f(x)$ is decreasing on $[a,b]$ / (a,b).

(iii) If $f'(x) \geq 0 \ \forall x \in (a,b)$, $f(x)$ is constant on $[a,b]$

Critical & Stationary point (The points where max/min exists)

Critical point: $f'(x_0) = 0$ / $f(x)$ is not differentiable at x_0 / $f(x)$ is not defined.



A, B, C, D are critical points.

Stationary point: $f'(x_0) = 0$ $\longrightarrow$ only.

A & B are stationary points.

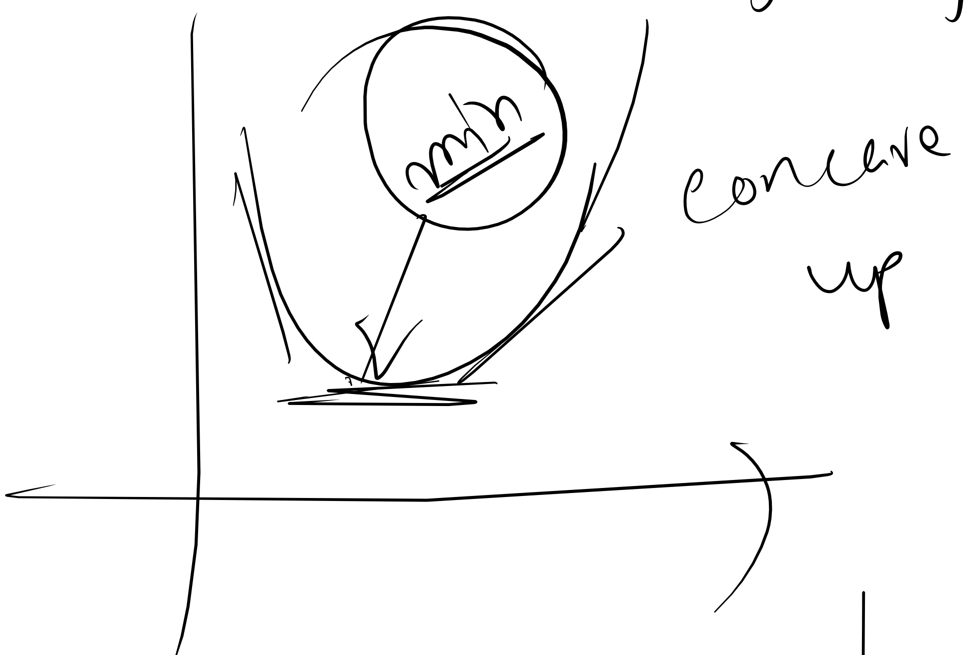
Concavity: let f be twice differentiable on an open interval (a,b).

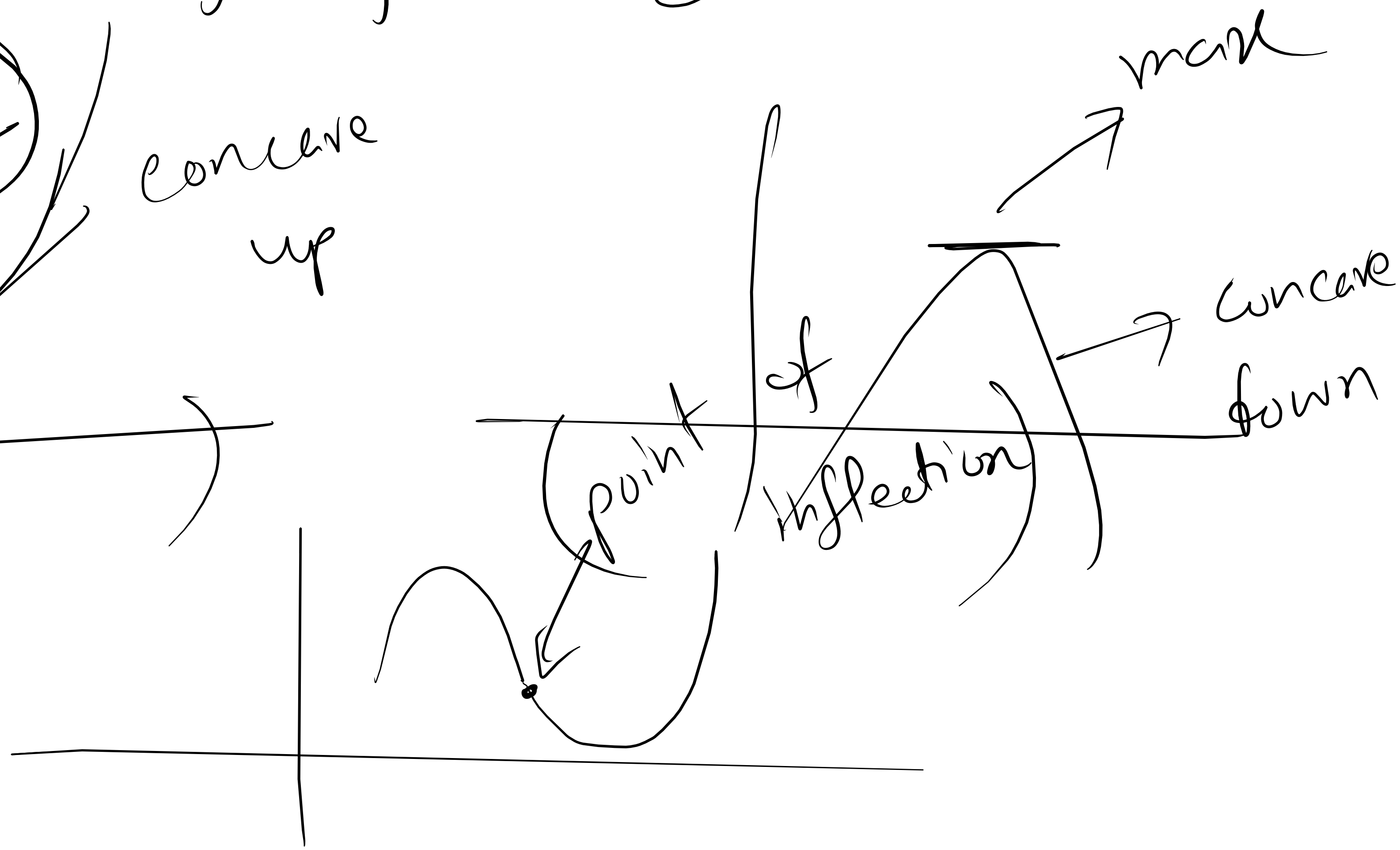
(i) If $\boxed{f''(x) > 0}$, $\forall x \in (a,b)$, f is concave up on $[a,b]$ / (a,b).

(ii) If $\boxed{f''(x) < 0}$, $\forall x \in (a,b)$, f is concave down on $[a,b]$ / (a,b).

(iii) $f''(x_0) = 0$, f has an inflection point at x_0

point of change of concavity (point of inflection)





$c \rightarrow$ critical points / stationary points.

<u>Mini</u> $\quad f''(c) > 0$, $\rightarrow$ $f(c)$ $\rightarrow$ minimum $\rightarrow$ point

$\rightarrow$ minimum.

c is a minimum domain point

<u>Maxi</u> $\quad f''(c) < 0 \longrightarrow$ maximum

$\downarrow$ c is a maximum domain point

$f(c)$ is maximum point.